Statistical Decision Making
45-752

Slide Set 2
Part 1

Topics covered in this slide deck:
Simple Regression
A/B Testing

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Simple Regression

In the Prob-Stat course, you studied simple regression.

- We begin with a brief review to refresh your memory.
- This includes an application.

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Application: Diamond Price and Weight

As our first example, we will use a dataset for diamonds.

The data are in file Diamonds_17.xlsx

Our sample consists of 17 diamonds that vary in weight (Carat).

All 17 diamonds have the same color, cut, and clarity.

The file contains the Price for which each diamond sold.

There is also a variable in the file named Observation which is simply a sequence of integers from 1 to 17. As we will see, this variable is convenient for referring to a diamond in the data set.

The following Cell in the accompanying notebook imports the data sets used in this slide deck.

```
# import Data
Diamonds_17_data = pandas.read_excel("data/Diamonds_17.xlsx")
Water_Conservation_Data=pandas.read_excel("data/Water_Conservation_Data.xlsx")
```


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Application: Diamond Price and Weight

To begin to get a feel for the data, view some summary statistics.

Run the following command in Colab

Diamonds_17_data.describe()

We see the following:

The average price is $3,523 and average carat weight is .756.
The diamonds range in weight from .69 to .85 carats.
The standard deviations are 247.27 and .044 for Price and Carat respectively.

	Observation	Price	Carat
count	17.000000	17.000000	17.000000
mean	9.000000	3523.117647	0.755882
std	5.049752	247.270066	0.044025
min	1.000000	2998.000000	0.690000
25%	5.000000	3331.000000	0.730000
50%	9.000000	3544.000000	0.760000
75%	13.000000	3642.000000	0.780000
max	17.000000	4085.000000	0.850000

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Simple Regression

Population model:

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

The intercept and slope parameters, β_0 and β_1 are unknown parameters.

The random component is denoted ϵ_i.

Ideal conditions for least squares are:

1. The ϵ_i are distributed independently of the x_i.
2. The ϵ_i are from a population with a mean of zero and a constant variance σ^2: $E(\epsilon_i) = 0$, $Var(\epsilon_i) = \sigma^2$. The standard deviation is an unknown parameter.
3. The error terms are not correlated with each other: $Cov(\epsilon_j,\epsilon_i)=0$ for all $i \neq j$.

Don't worry too much about these for the time being. They will become clearer as we go along.

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Least Squares Estimation

We choose estimators of the population parameters β_0 and β_1 to minimize the sum of the squared estimated error terms.

We denote the estimators b_0 and b_1.

In Mini 1, you saw that the Least-Squares Coefficient Estimators are:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1\,\bar{x}$$

The notational distinction is important! Memorizing formulas is not!

- β_0 and β_1 are the unknown population coefficients.
- b_0 and b_1 are the estimates we calculate from our data.
- The estimated variance of the error term is given by $s_e^2 = \frac{\sum e_i^2}{n-2}$
- We divide by (n-2) because we must estimate two coefficients, b_0 and b_1, before we can calculate the estimated errors, e_i. This is a "degrees of freedom" correction.

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Regression with Diamond Data and Formulas for Coefficients

Using the Python script, run the regression. I've named it OurReg

OurReg = tprstats.model("cs", "Price~Carat", data=Diamonds_17_data)

View the results with robust std errors (explained in slides that follow):

OurReg.summary()

OLS Regression Results

Dep. Variable:	Price	R-squared:	0.588
Model:	OLS	Adj. R-squared:	0.560
No. Observations:	17	F-statistic:	12.32
Covariance Type:	HC1	Prob (F-statistic):	0.00316

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	268.7373	927.890	0.290	0.776	-1709.013	2246.488
Carat	4305.4059	1226.850	3.509	0.003	1690.438	6920.374

$$b_0 = \bar{y} - b_1\,\bar{x}$$

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

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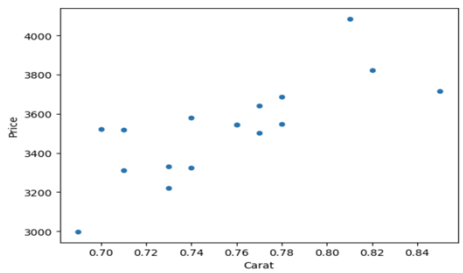
Scatter Plot of Data

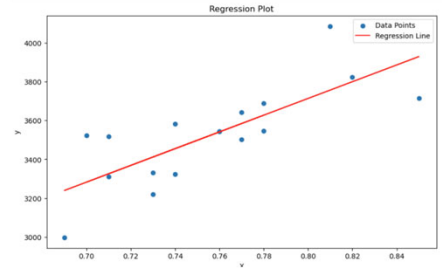
Using the Python script do a scatterplot of Price vs Carat

```
#Do a Scatterplot of Price versus Carat

Diamond_17_plot = Diamonds_17_data.plot.scatter(x="Carat", y= "Price")
Diamond_17_plot
```



Add Regression Line and Label The Plot

```
# Add the regression line
# Predicted values
y_pred = OurReg.predict()

# Plotting
plt.figure(figsize=(10, 6))
# drops the constant term model.X.iloc[: , 1:]
plt.scatter(OurReg.X.iloc[: , 1:], OurReg.y, label='Data Points')
plt.plot(OurReg.X.iloc[: , 1:], y_pred, color='red', label='Regression Line')
plt.xlabel('X')
plt.ylabel('y')
plt.title('Regression Plot')
plt.legend()
plt.show()
```



Don't worry about the details of commands for making this plot. I won't ask you to do this on a quiz or final exam.
The objective is for you to see how the regression line appears relative to the data.

Coefficient Interpretation

Variable	Coefficient
Carat	4,305.41
Constant	268.737

Here is how to interpret the coefficients:

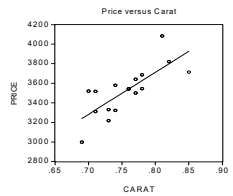
Carat (Slope): "For every .1 unit increase in Carat, price is predicted to increase by $430.54."

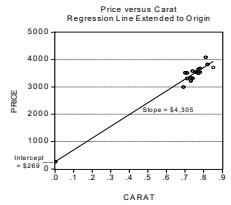
Constant (Intercept): "When Carat=0, predicted price is $268."

Why is the estimated intercept not equal to zero?
This typically happens when we have no data close to x=0. The intercept is then an extrapolation of the regression line to x=0. In such cases, we typically do not attach much importance to the value of the intercept.

Note, in this application, that the value for the intercept is not significantly different from zero. Even if it were significant (or even negative), we would probably not be too concerned about it in this application. The next page illustrates why.

Illustration of Extrapolation to Carat=0





Both plots show the same data:
- The top scatter plot showcases the data range and does not include the origin.
- The bottom scatter plot includes the origin.

In this application, we extrapolate a long way from the data to the point where the regression line intercepts the vertical axis.
- In such situations, the estimated intercept is often not very meaningful.

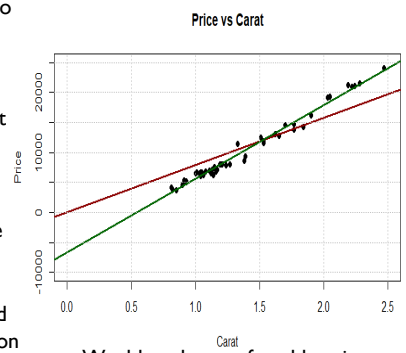
Why Not Force Intercept to Equal Zero?

So, why not force the intercept to zero?

The example at right has larger diamonds. Allowing a non-zero intercept (green) permits us to fit the data much better than if we force the intercept to zero (red line).

This is why it is standard practice to always have an intercept in a regression.

When we do prediction, we need to keep in mind that the prediction becomes problematic if we extrapolate far from the range of our data.

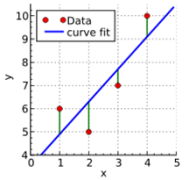


Would we be comfortable using our regression to predict price of this diamond

Why Minimize Sum of Squared Errors?

Why minimize sum of squared estimated error terms? Real-world data will not fit a regression line perfectly. (It would be trivial to fit the line if they did).

- To find the line that "best fits" the data points, we might consider minimizing the sum of the residuals – the sum of the distances between the line and the points. This is not a sensible procedure. There are endless lines that satisfy this criterion, with positive residuals offsetting negative residuals so that the sum equals zero.
- We might consider choosing the line that minimizes the sum of the absolute value of the residuals. This is called "Least Absolute Deviation" or LAD. In the early days, this was computationally infeasible. It can now be done easily. This method is not used very much in practice.
- We can choose the line that minimizes the sum of squared residuals. This is the standard method used in practice. It is computationally very simple—obtained by solving a system of linear equations.
- More importantly, the least squares estimators of the coefficients have nice properties, as proved in the famous Gauss-Markov Theorem. This theorem says that, if certain conditions are satisfied, least squares is the Best Linear Unbiased Estimator of the coefficients in the linear regression model.



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Simple Regression: Hypothesis Testing

Standard Errors of Coefficients

Loosely speaking, the standard error of a coefficient gives us a measure of accuracy of the coefficient estimate.

- We use the standard error of a coefficient when we test a hypothesis about a coefficient.
- We also use the standard error of a coefficient to form a confidence interval for the coefficient.

Standard Errors of Coefficients

There are two ways of calculating standard errors of coefficients.

- Old-Fashioned Approach: This approach assumes that the variability of the errors is roughly the same across the range of values of the right-hand-side variable. This approach yields biased estimates of standard errors if this assumption is not satisfied.

  The optional appendix derives standard errors using old-fashioned approach.

- Modern approach (Robust Standard Errors): This approach does not assume that the variability of the errors is the same across the range of values of the right-hand-side variable. The standard errors are valid regardless of whether heteroskedasticity is present. Hence, they are called "Robust" standard errors.

  The optional appendix derives robust standard errors

Heteroskedasticity definition: When the variability of the residuals is not the same across the values of the right-hand-side variable, we say that the residuals exhibit heteroskedasticity.

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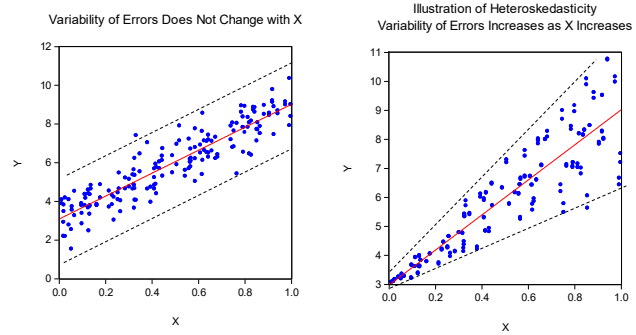
Heteroskedasticity Illustrated

Left panel illustrates a case in which variability of residuals is roughly constant across the range of X (i.e., no heteroskedasticity).

Right panel illustrates a case in which variability of residuals increases with X

When the variability of the residuals changes as X changes, this is called Heteroskedasticity.



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Robust Standard Errors

When heteroskedasticity is present:

- Coefficient estimates are unbiased. Good!
- Standard formula for calculating standard errors of coefficients is biased, even in large samples. Not so good!
- The tprstats command that we used earlier in these slides calculates standard errors that are valid even if heteroskedasticity is present. We will use the tprstats command as in this slide deck to view regression output except for models with time series data.
- Later in the course, we will introduce a variant of the command that we use for time series data.

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Comparison of Standard Errors for Diamond Regression

Below are the results with the two approaches to calculating standard errors, bordered in red. Note that the standard errors, shown in parentheses, and the t-statistics and p-values all differ.

	Dependent variable:	
	Price	
	Old-fashioned (1)	Robust (2)
Carat	4,305.406***	4,305.406***
	(931.269)	(1,226.850)
	t = 4.623	t = 3.509
	p = 0.0004	p = 0.0005
Constant	268.737	268.737
	(705.052)	(927.890)
	t = 0.381	t = 0.290
	p = 0.709	p = 0.773
Observations	17	17
R2	0.588	0.588
Adjusted R2	0.560	0.560
Residual Std. Error (df = 15)	163.998	163.998
F Statistic (df = 1; 15)	21.374***	21.374***
Note:	*p<0.1; **p<0.05; ***p<0.01	

The coefficients are the same for both approaches.

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Hypothesis Testing

Application: Test the null hypothesis that the population slope coefficient in the diamond model equals 0. Use a 5% significance level (i.e., α=.05):

H_0: $\beta_{1,0} = 0$

H_1: $\beta_{1,0} \neq 0$

From the regression output, we see that the slope coefficient is b_1=4,305 and the standard error of the slope coefficient s_{b_1}= 1227.

Our test statistic is: t = $\frac{b_1-\beta_{1,0}}{s_{b_1}} = \frac{4305-0}{1227} = 3.509$

Using : 2*t.cdf(-3.509,15) , we find that the p-value =.003.

Our p-value < α, so we reject the null hypothesis that the slope coefficient = equals 0.

Look back to the regression output and you will see that Python calculates the above t-statistic and p-value.

The information to test the null hypothesis that a coefficient equals zero is always available in the Python output.

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Hypothesis Tests in Regression Output

The results below are from the Python regression output presented earlier in these slides.

Note the t-statistic and p-value for the coefficient of Carat from the Python output are the same as the values we calculate on the previous slide:

	Estimate	Std. Error	t value	P>|t|
(Intercept)	268.74	927.89	0.290	0.776
Carat	4305.41	1226.85	3.509	0.003

Statistical software packages generally test the null hypothesis that the population coefficient of each variable equals zero. Why?

- When we create a regression model, we generally think that our right-hand-side variable has predictive value.
- If we reject the hypothesis that the population coefficient equals zero, we then have evidence that the right-hand-side variable does indeed have predictive value.

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Confidence Interval for A Coefficient

Confidence Interval for a Coefficient: Let β_1 be the unknown population coefficient for Carat. We use the estimated coefficient of Carat, b_1, and its estimated standard error, s_{b_1}, to form the confidence interval.

We form an interval that contains β_1 with probability (1-α).

$$[b_1 - t_{\nu,\alpha/2} s_{b_1}, b_1 + t_{\nu,\alpha/2} s_{b_1}]$$

The number of degrees of freedom equals the number of observations minus the number of coefficients we estimated: ν = n-2 = 17-2 = 15.

Recall that we calculate $t_{\nu,\alpha/2}$ in Python using t.ppf(1- α/2, ν)

Applying the above, we obtain the following confidence interval for the coefficient of carat:

[4305.41- 2.1314*1226.85, 4305.41+2.1314*1226.85] = [1690, 6920]

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Confidence Intervals for Coefficients

Interpretation: There is a 95% probability that the interval [1690,6920] contains the population slope coefficient β_1.

This is a very wide confidence interval. A larger sample would be helpful in improving precision of the estimated coefficient.

The Python output is repeated below so that you see where the confidence intervals of the coefficients are provided.

OLS Regression Results

Dep. Variable:	Price	**R-squared:**	0.588
Model:	OLS	**Adj. R-squared:**	0.560
No. Observations:	17	**F-statistic:**	12.32
Covariance Type:	HC1	**Prob (F-statistic):**	0.00316

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	268.7373	927.890	0.290	0.776	-1709.013	2246.488
Carat	4305.4059	1226.850	3.509	0.003	1690.438	6920.374

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Simple Regression: Fitted Values, R^2, and Prediction

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Fitted and Predicted Values

The predicted value of y for a given x_i is given by: $\hat{y}_i = b_0 + b_1 x_i$

It is very common to use a "hat" ^ above a variable to denote a predicted value, as we have done above.

Definition: The predicted values of y with the x values used in estimating the regression the are called fitted values.

In Python, we calculate fitted values using the command below. On the right-hand side, you put the name of your regression before the period.
yhat=OurReg.predict()

I named the fitted values yhat, but you can choose any name you wish.

In the accompanying script, you can view the yhat values.

- You will see that the fitted price, yhat, for diamond 13 is $3,584.
- In the Excel data file, you will see that the price for diamond 13 is $3,501.
- For this diamond, the regression model over-predicts price by $83.

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Estimated Errors

The estimated error terms in a regression model are called residuals.

The estimated error term for an observation is the difference between the actual value y_i and the fitted value $\hat{y}_i$

For example, on the previous page, we saw that for diamond 13 (i=13), the difference between the actual and fitted value was $y_i - \hat{y}_i = -\$83$.

We use the symbol e_i to denote the estimated error term for observation i: $e_i = y_i - \hat{y}_i$

We calculate and view the residuals in Python using the two commands shown below using the name you choose for your regression. As you can see, I named the result resid.

```
#Calculate the residuals from the regression and name them resid
resid=OurReg.result.resid
resid
```

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Residuals from Diamond Regression

The commands on the previous slide display the residuals in Colab as shown on the right.

As you see, Colab numbers the residuals 0 to 16 instead of 1 to 17.

If we use the standard numbering of observations, we obtain the numbers on the far right panel.

We then see that the residual for observation 13 is -83, as we determined on the previous slides.

Any time you view residuals with Colab, you will want to recall the peculiar numbering convention of Colab.

		Observation
0	-80.683612	1
1	3.154211	2
2	239.478566	3
3	-14.575493	4
4	61.046093	5
5	328.883915	6
6	23.829856	7
7	-213.332322	8
8	-241.467375	9
9	58.100152	10
10	-79.953907	11
11	-191.683612	12
12	-82.899848	13
13	126.262329	14
14	3.154211	15
15	191.424507	16
16	-130.737671	17

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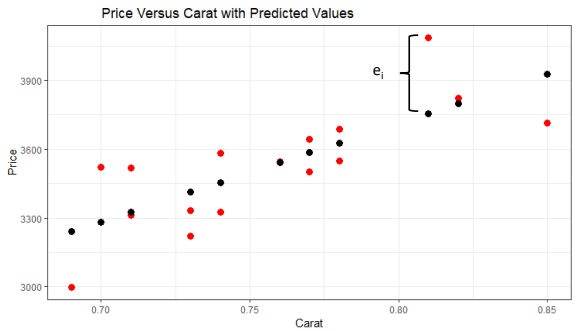
Scatter Plot of Actual, Fitted, and Residuals

This graph shows actual prices for which the diamonds sold (red) and fitted prices (black). Both are plotted versus Carat.

The vertical differences between actual and fitted are the estimated error terms, e_i, which, as note previously, are also called residuals.



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R^2 Values

The R^2 statistic is often referred to as the proportion of variance in y that is "explained" by the right-hand-side variable(s) in the regression.

The "total sum of squares of y can be separated into the "explained" by the predicted values, $\hat{y}_i$, and the remaining "unexplained" variation in y.

$$\underbrace{\boxed{\sum_{i=1}^{n}(y_i-\bar{y})^2}}_{\text{Total}} = \underbrace{\boxed{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}}_{\text{Explained}} + \underbrace{\boxed{\sum_{i=1}^{n}e_i^2}}_{\text{Unexplained}}$$

The R^2 statistic is the ratio of the "explained" to the "total" = $\dfrac{\sum_{i=1}^{n}(\hat{y}_i-\bar{y})^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}$

The R^2 lies between 0 and 1.

A value of 1 would be an exact fit of the data to the regression line.

A value of 0 would be a horizontal line—x would have no value in predicting y.

The next slide illustrates R^2. All five regressions have the same intercept and slope, and range of the axes is same for all graphs.

The R^2 statistic is a useful summary measure but it tends to be over-emphasized in practice. We will discuss this in a future note set.

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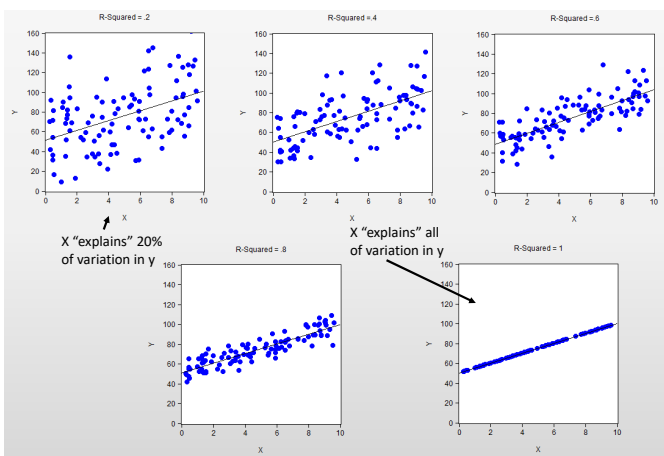
Illustration of R^2



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Predicting Value of a New Diamond

Application: A customer is considering purchase of a new diamond that has weight = .75 carats. The customer asks us to predict the price that he can expect to pay and also to form a 95% prediction interval.

Using our regression equation, which I named OurReg:

```
#Predicted price and prediction interval
# for a diamond with carat=.75
Xnew = pandas.DataFrame({'Carat': [0.75]})
OurReg.prediction_intervals(Xnew)
```

We obtain:

Predicted	Lower	Upper
3497.79173	3137.804541	3857.778919

Thus, if we were considering purchase of a .75 carat diamond:

- Our predicted price would be $3,498
- With 95% probability the price would be between [$3,138, $3,858]

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Prediction for Several New Diamonds

Suppose a jeweler obtains three new diamonds.
The diamonds have Carat weights .62, .74, .81.

Calculate predicted values and prediction intervals with the following commands:

Note in the command that [.62,.74,.81] puts the three values in a vector.

```
#Predicted price and prediction interval for three diamonds
Xnew = pandas.DataFrame({'Carat': [.62,.74,.81]})
OurReg.prediction_intervals(Xnew)
```

We obtain the following predicted values and prediction intervals:

Predicted	Lower	Upper
2938.088961	2432.962197	3443.215724
3454.737671	3092.738025	3816.737317
3756.116085	3369.343783	4142.888387

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Prediction Practice

Suppose a jeweler obtains three new diamonds with carat weights .69, .77, and .88 respectively.

Predict the prices of these diamonds and form a 95% prediction interval for each.

You should obtain:

Predicted	Lower	Upper
3239.467375	2840.941651	3637.993099
3583.899848	3222.254060	3945.545637
4057.494499	3572.566828	4542.422171

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Some Useful Terminology

In regression analysis:

Sample: The set of data that we use for our regression is called our "sample".

Observation: We call a particular data point an "observation".

- An observation is a row of data in our spreadsheet.
- We often refer to an observation by a number. For example, observation 13 in our spreadsheet for the diamond data contains the price and carat weight of the 13th diamond in the spreadsheet.
- Sometimes, we use index i to denote an observation. In our current application we have 17 diamonds. Hence, index i can take on a value between 1 and 17.

Cross Section Data: Our diamond data are for several different diamonds sold over a short period of time. In general, a cross section contains observations obtained at roughly the same point in time. If we gathered CEO pay for a given year for a sample of companies, it would be a cross section.

Time Series Data: A sequence taken at successive points in time. Examples: Daily returns on the SP500 comprise a time series. A series with monthly data for inflation in the US is a time series.

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Diagnostics

- When you have estimated a regression, it is a good idea to do some diagnostics to investigate the model.
- One useful diagnostic is to plot the residuals to see if there are outliers or other unexpected patterns.
- Another useful diagnostic is to plot the actual values of the dependent variable along with the fitted values and prediction intervals.

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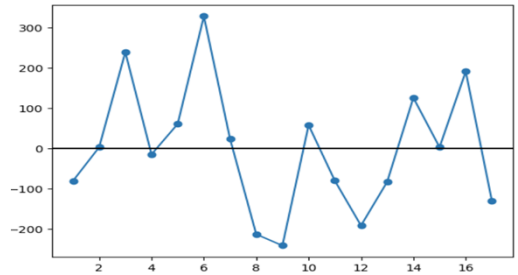
Plotting Residuals

Run the commands in the Colab script to plot the residuals for our 17 diamonds.

Note that observation number appears on the horizontal axis for each diamond. This is the row in the data spreadsheet where the diamond appears.

From the plot, we see that there are no major outliers or other patterns that might suggest the model is not suitable.

We will see later in the course examples of patterns that suggest need to change some aspect of the regression analysis.



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Another Useful Diagnostic Plot

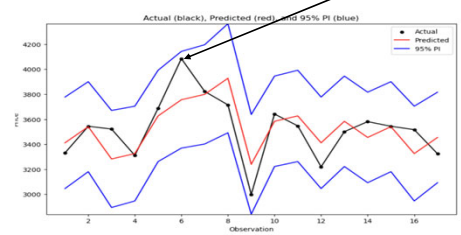
Plot the actual prices, the fitted values, and the residuals and prediction interval in a single graph for all 17 diamonds

```
# Plot actual prices (black dots), fitted values (red), and 95% prediction intervals (blue)
# Regression name, OurReg
OurReg.plot_actual_fitted()
```

The observation number for each diamond appears on the horizontal axis.

We expect about 5% of the actual prices to lie outside their prediction intervals.

In our small sample, none lie outside the prediction interval, but one is close.



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Summary of Review of Regression

Our review of simple regression has covered:

- Simple regression population model
- Estimating simple regression model using Python
- Interpreting coefficients
- Definition of estimated error terms
- Calculating fitted values
- Interpreting R^2 statistic
- Testing a hypothesis about a coefficient
- Prediction and Prediction Intervals
- Some useful diagnostics

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Statistical Decision Making
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A/B Testing

This file is named: Stat_Mini4_Slide_Set_2_2025.pptx
The accompanying Python notebook is:
Stat_Mini4_Script_Slide_Set_2_2025.ipynb

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A/B Testing in Business

As noted in slide set 1, there has been an explosion in the use of A/B testing in industry in recent years. All of us here have likely participated in such A/B testing. Here is a prompt and response from Copilot.

Prompt: Tell me about A/B testing in large companies using 100 words or less.

Response:

A/B testing is a crucial practice for large enterprises. It involves comparing two versions of a webpage or app feature (Version A and Version B) to determine which performs better. Companies like Airbnb, Amazon, Google, and Microsoft use A/B testing extensively. For instance, Airbnb optimizes search rankings, Amazon tests website layouts, Google refines ad content, and Microsoft experiments with product features.

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Use of A/B Testing by Amazon

Quote from Amazon founder Jeff Bezos, "Our success at Amazon is a function of the number of experiments we run per year, per month, per day."

Prompt to Copilot: Explain how Amazon uses brand registry in A/B testing using 100 words or less.

Copilot Response: Amazon uses its Brand Registry to facilitate A/B testing through the "Manage Your Experiments" tool. This tool allows brand-registered sellers to test different versions of product content, such as titles, images, and descriptions, by showing them to different customer segments. The results help sellers identify which version performs better in terms of customer engagement and sales, enabling data-driven decisions to optimize product listings and increase conversions. https://sell.amazon.com/tools/manage-your-experiments

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A/B Testing —Some Background

Experiments are extensively used in medical research.

Individuals with an illness to be treated by a new medication are randomly assigned to one of the following groups:

Treatment Group: Those who receive the new medication.

Control Group: Those who receive the best existing alternative.

Randomization assures that the severity of an individual's condition does not affect the group to which the individual is assigned.

Where feasible, experiments are double-blind.

- Participants in the study do not know if they are in the treatment or control group.
- The individual dispensing the medication does not know which participants are in the treatment group and which are in the control group.

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Background on Clinical Trials in Medicine

The 1938 Food, Drug, and Cosmetic Act required drug sponsors to submit safety data to US Food and Drug Administration (FDA) officials for evaluation prior to marketing.

Despite this law, it is estimated that only 45% of drugs were tested against a control group in 1950.

Beginning in 1958, hearings on the drug industry were held by Senator Estes Kefauver. These hearings drew attention to the poor state of clinical trial research as it had been conducted (or failed to be conducted) under the 1938 statute.

The Kefauver-Harris 1962 amendments to the 1938 law required that future drugs would have to be demonstrated both save and effective. This provision required controlled trials to support claims of efficacy. The provision mandated A/B testing for drug approval.

https://www.fda.gov/media/110437/download

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A Patient In Need of Treatment

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Statistical Decision Making

Testing A Difference in Population Means

Statistical Framework and an Application

Book Chapter 3 provides further reading on this topic
03_Testing_Difference_in_Population_Means.pdf

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Water Conservation Experiment

We now study an experiment utilizing A/B testing.

We will lay out framework for A/B testing and analysis of the results of the experiment.

The Cobb County Water System (CCWS) serves residential customers in metropolitan Atlanta, Georgia.

- Atlanta was experiencing drought conditions, making it difficult for CCWS to provide enough water for all of its customers.
- CCWS conducted an experiment to encourage water conservation.

This is an exceptionally well designed and implemented experiment!

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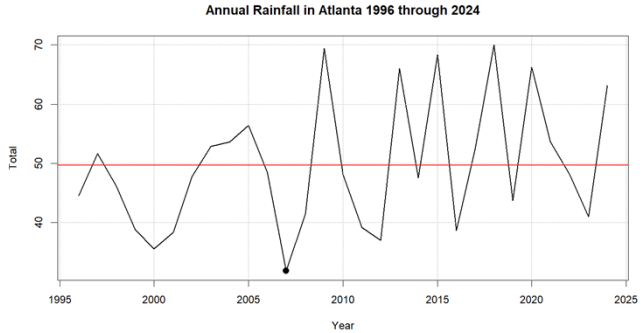
Annual Rainfall Atlanta Since 1996

Atlanta rainfall in 2007 was the lowest rainfall over the period 1996 through 2024.



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Water Conservation Experiment Continued

In the spring of 2007, each household was randomly assigned to one of four groups, a Control Group and three different treatment groups.

Households in the control group did not receive any mailing.

Household in the different treatment groups received the following:

- Treat1: "Tip sheet", information only
- Treat2: Augment "tip sheet" with personal plea to reduce usage
- Treat3: Explicit comparison of water usage to other Atlanta residents, as shown on next slide

Water usage data was then obtained for each household for each summer from 2007 through 2013.

Then average water usage of those in the control group was compared to average usage of those in each treatment group.

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Example of Treatment 3 Letter

As we enter the summer months, we thought that you might be interested in the following information about your water consumption last year:

Your own total consumption June to October 2006: 52,000 gallons

Your neighbors' average (median) consumption June to October 2006: 35,000 gallons

You consumed more water than 73 percent of your Cobb County neighbors.

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You Might Be Wondering…

The description on the previous slide might leave you wondering if this involves A/B testing given that there are three treatment groups and a control group.

In actuality, the researchers conducted three A/B tests.

- Treatment 1 versus Control
- Treatment 2 versus Control
- Treatment 3 versus Control

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Testing Difference in Population Means

Testing of a differences in means lies at the heart of A/B testing. The water conservation experiment provides a nice application.

The experiment summarized on the previous slide is described in the case write-up posted for you for this session. In the file, it is named Water_Conservation_Case_aer.103.1.pdf.

The experiment is the work of:
Paul J. Ferraro, Juan Jose Miranda, and Michael K. Price, "The Persistence of Treatment Effects with Norm-Based Policy Instruments: Evidence from a Randomized Environmental Policy Experiment".
http://www.aeaweb.org/articles.php?doi=10.1257/aer.101.3.318

I thank these authors for their generosity in sharing their data with us.

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Hypothesis To Be Tested

Consider comparison of Treatment Group 3 and the Control Group

Population: Current and future CCWS customers

μ_X and σ_X : Mean and standard deviation of water consumption by the population if a social comparison message is sent to all.

μ_Y and σ_Y : Mean and standard deviation of water consumption by the population if no letter is sent to any customer.

Null hypothesis: We use subscript 0 to denote the null hypothesis

Social Comparison letter will have no effect: $\mu_{X,0} = \mu_{Y,0}$

Alternative hypothesis:

Social Comparison letter will have an effect: $\mu_{X,0} \neq \mu_{Y,0}$

Significance level α =.05

There is a 5% probability we will reject the null hypothesis when it is true.

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The Experiment: Treatment 3 vs Control

Treatment Group 3: 11,675 households were randomly selected to receive the Social Comparison letter

Control Group: 71,643 household were randomly selected to not receive a letter.

Letters were sent to each household in each treatment group in May 2007.

Water consumption of each household was measured for each summer from 2007 through 2013.

In these slides we will study the effect on water usage in the summer of 2007. Later we will study whether effects persisted in subsequent years.

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Results of The Experiment

In Summer 2007: Households in the treatment group consumed, on average, 1,614 gallons less water than households in the control group.

Was this caused by Treatment 3, or could it be due to chance?

Our hypothesis test will help us judge whether the treatment caused the reduction in water use.

We now lay out the framework for the hypothesis test:

Water consumption in Summer 2007 in thousands of gallons:

$\overline{X}$ = Average water consumption in treatment group = 34.858

$\overline{Y}$ = Average water consumption in control group = 36.472

$\overline{X}$ - $\overline{Y}$ = Treatment Effect = 34.858-36.472 = -1.614

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Calculating Standard Error of Estimated Treatment Effect

Random sampling implies: $\mathrm{Var}(\overline{X}-\overline{Y}) = \frac{\sigma_X^2}{n_x} + \frac{\sigma_Y^2}{n_Y}$

We estimate σ_X^2 and σ_Y^2 using :

$$s_X^2 = \frac{\sum_{i=1}^{n_X}(X_i-\overline{X})^2}{n_X-1} = 693.53 \text{ and } s_Y^2 = \frac{\sum_{i=1}^{n_Y}(Y_i-\overline{Y})^2}{n_Y-1} = 855.68$$

We then calculate the estimated standard error of ($\overline{X}$ - $\overline{Y}$)

$$\sqrt{\frac{s_X^2}{n_x} + \frac{s_Y^2}{n_Y}} = .267$$

This standard error measures the accuracy of the treatment effect.

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Test Statistic Continued

Under our null hypothesis, $(\mu_{X,0} - \mu_{Y,0})=0$ the t-statistic is the ratio of the estimated treatment effect to the estimated standard error:

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_{X,0} - \mu_{Y,0})}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}} = \frac{(\bar{X} - \bar{Y}) - 0}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}}$$

This statistic is the workhorse of A/B testing.

Substituting the information from the previous slides, we obtain our calculated t statistic:

$$t = \frac{-1.614}{.267} = -6.04$$

This tells us that our estimated treatment effect is 6 times larger than the estimated standard error of the treatment effect. back

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Completing the Hypothesis Test Using The p-value

Our number of degrees of freedom is:
$(n_X+n_Y-2)=(11{,}675+71{,}643-2) = 83{,}317$
In Python, we calculate the p-value using: *2 * t.cdf(-6.04,83317)*
We obtain a very small p-value = $1.5\times10^{-9} \approx 0$
The p-value < α. We reject the null hypothesis and conclude that the treatment (letter) had a significant effect on water consumption. This is good news for CCWS!
Imagine the celebration that these researchers did when they obtained this result!
Interpretation of the p-value: The p-value is the probability of obtaining a treatment effect as big or bigger than 1.614 if the null hypothesis is true.
Logic: A treatment effect as big as 1,614 gallons per household would be extraordinarily unlikely if the null hypothesis is true. Hence, we conclude the null is false and thereby conclude that the experiment caused reduction in water use.

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Statistical Decision Making

Testing A Difference in Population Means
Using Regression

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Using Regression to Estimate and Test A Difference in Means

- In the previous slides, we worked through the statistical theory for testing a difference in means.
- We then worked through an application doing the calculations "by hand" for the water conservation case.
- It is important for you to know how to the calculations by hand for situations in which you know sample means and sample standard deviations but you do not have the data.
- When you have the data, regression provides a quick and easy way to estimate a difference in population means and to conduct a test of a difference in population means.

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Import Water Conservation Data

We now see how to use regression to evaluate the effect of treatment 3.

Import file Water_Conservation_Data.xlsx

Variable GROUP denotes the experimental assignment of each household:

- Households in GROUP 1 received treatment 1.
- Households in GROUP 2 received treatment 2.
- Households in GROUP 3 received treatment 3.
- Households in GROUP 4 were in the control group.

We will use regression to compare the outcomes in Summer 2007 for households in the treatment 3 group to households in the control group.

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Select Data for A/B test

Select households that were in either GROUP 3 or GROUP 4.

```
# Select the observations in treatment Group 3 and the control group, Group 4
WatDat34 = Water_Conservation_Data.query('GROUP==3|GROUP==4')
```

In the data file:

- Variable SUMMER_07 contains water use of each household in the summer of 2007.
- Variable TREAT3 equals 1 for households who received treatment 3 and equals 0 otherwise.
- A variable such as TREAT3 that takes on values of 0 or 1 is called an indicator variable.

Run regression to estimate effect of treatment 3 and view the regression output. As you see, I named the regression WatReg34.

```
# Run regression to estimate effect of treatment 3 and view the regression output
WatReg34 = tprstats.model("cs", 'SUMMER_07~TREAT3', WatDat34)
WatReg34.summary()
```

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Estimating Effectiveness of Treatment 3

Mean water use in the control group was 36,473 gallons

Difference between treatment 3 and control group was -1614 gallons

Is this treatment effect significant at the 5% level? The 1% level?

Dep. Variable:	SUMMER_07	R-squared:	0.000
Model:	OLS	Adj. R-squared:	0.000
No. Observations:	83319	F-statistic:	36.54
Covariance Type:	HC1	Prob (F-statistic):	1.50e-09

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	36.4725	0.109	333.730	0.000	36.258	36.687
TREAT3	-1.6145	0.267	-6.045	0.000	-2.138	-1.091

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Look back and you will see that the first four values in the row for TREAT3 are the values we calculated by hand when we analyzed the data.

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Derivation to Clarify Interpretation of the Regression Coefficient

We now demonstrate that the coefficient of TREAT3 is indeed the effect of Treatment 3.

From the previous slide the estimated regression is: $\hat{y}_i = 36.472 - 1.614 * TREAT3_i$

We now work through interpretation of the regression coefficients.

For all households in the control group: $TREAT3_i = 0$

Hence, for households in the control group: $\hat{y}_c = 36.472 - 1.614 * 0 = 36.472$

For all households in the treatment group 3: $TREAT3_i = 1$

Hence, for households in the treatment 3 group: $\hat{y}_{T3} = 36.472 - 1.614 * 1 = 34.858$

The difference between treatment 3 and the control group is then:

$$\hat{y}_{T3} - \hat{y}_c = 34.858 - 36.472 = -1.614$$

This accords with our interpretation on the previous slide.

- The intercept is mean water use in the control group, 36.472.
- The coefficient of TREAT3 if the treatment effect of treatment 3, -1.614.

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A/B Testing: Another Application

At the right, you will see a photo of the front of a box of Zicam.

You will see the assertion that Zicam helps with 6 symptoms.

If you look to the right of Symptoms, you will see the following footnote symbol‡

The footnote is in small print.

I looked on the Zicam web page and found that the footnote says:

‡ Claims based on traditional homeopathic practice, not accepted medical evidence. Not FDA evaluated.

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A/B Testing of Echinacea

You will see on the box on the previous page that Echinacea is one of the ingredients of Zicam.

There have been several clinical trials of Echniacea. The findings of these studies were reviewed and summarized in the study on the following web page: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4068831/

The objectives of the analysis and the authors' conclusions are reported below:

Objectives

To assess whether there is evidence that *Echinacea* preparations are effective and safe compared to placebo in the prevention and treatment of the common cold.

Authors' conclusions

Echinacea products have not here been shown to provide benefits for treating colds, although, it is possible there is a weak benefit from some *Echinacea* products: the results of individual prophylaxis trials consistently show positive (if non-significant) trends, although potential effects are of questionable clinical relevance.

In spite of this evidence of very limited effectiveness:

https://www.nasdaq.com/press-release/church-dwight-acquires-zicam-brand-for-$530-million-2020-12-01

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Summary Thoughts Regarding A/B Testing

- We have seen how to test a difference in means by hand, and by using regression.
- This approach is very straightforward to use and provides results that are easy to interpret.
- We will use this approach in several applications during our course.
- I think there is a high likelihood that you will find opportunities for using this approach in your work.
- For practice, use regression to conduct an A/B test to investigate the effectiveness of Treatment 2 of the Water Conservation Experiment relative to the Control Group. Just follow the steps that we did with Treatment 3.
  - You will find that Treatment 2 also significantly reduced water consumption relative to the control group.
  - You will also find that the effect of Treatment 2 is smaller than the effect of Treatment 3.

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Summary of First Two Sets of Slides

- We Reviewed some Fundamental Statistics Concepts
  - Central Limit Theorem
  - Hypothesis Testing
- We Reviewed Statistical Quality Control
- We Reviewed Testing Difference in Means and Applied to Water Conservation Experiment
- We Reviewed Single Variable Regression
- We Saw How to Use Regression to Test Difference in Means and Illustrated with Water Conservation Application
- We saw how to use Python to utilize the statistical methods we have developed.

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Optional Appendices

- The following appendices provide more detailed information about topics in the text of these slides.
- You are not required to be familiar with these materials, but you may find them interesting.

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Deriving the Variance of b_1 When Ideal Conditions Hold 1 of 2

Our model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

The variance of y_i given x_i: $Var(y_i) = Var(\beta_0 + \beta_1 x_i + \varepsilon_i) = Var(\varepsilon_i) = \sigma^2$

Rewrite b_1 as follows:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{\sum(x_i - \bar{x})y_i}{\sum(x_i - \bar{x})^2} - \frac{\bar{y}\sum(x_i - \bar{x})}{\sum(x_i - \bar{x})^2} = \sum_{i=1}^{n}\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)y_i$$

The term in red font equals zero because the sum of deviations of any variable from its mean equals zero.

The next page details derivation of variance of b_1. Color coding of the = signs are used in explanation of each step.

The derivation uses the ideal conditions listed near the beginning of this note set.

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Deriving the Variance of b_1 When Ideal Conditions Hold 2 of 2

Given the x values, the variance of b_1 is:

$$Var(b_1) = Var\left\{\sum_{i=1}^{n}\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)y_i\right\} = \sum_{i=1}^{n} Var\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)y_i\right\}$$

$$= \sum_{i=1}^{n}\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)^2 Var(y_i)\right\} = \sum_{i=1}^{n}\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)^2 \sigma^2\right\}$$

$$= \frac{\sigma^2}{[\sum(x_i - \bar{x})^2]^2}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}$$

Explanation of each step above by the color of the = sign:
Red: The y_i are independent of each other because, under ideal conditions, the error terms are independent of each other. The x's are given (non-random). Hence, variance of sum equals sum of variances.
Green: The x's are non-random. The non-random part is squared and multiplied by variance of the y_i.
Blue: Under ideal assumptions, the variance of y_i given x_i is constant σ^2.
Magenta: Terms that do not vary with i are taken out of the summation.
Orange: Simplify by cancelling terms.
The standard error is the square root of the last term.

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Deriving the Variance of b_1 When There Is Heteroskedasticity

Given the x values, the variance of b_1 when there is heteroskedasticity is:

$$Var(b_1) = Var\left\{\sum_{i=1}^{n}\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)y_i\right\} = \sum_{i=1}^{n} Var\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)y_i\right\}$$

$$= \sum_{i=1}^{n}\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)^2 Var(y_i)\right\} = \sum_{i=1}^{n}\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)^2 Var(\varepsilon_i)\right\}$$

Note that steps above are the same as on previous slide except we do not make the substitution $Var(y_i) = \sigma^2$ because the variance of the errors is not constant when there is heteroskedasticity.

We do not know $Var(\varepsilon_i)$. We can obtain valid standard errors in large samples by using the square of the estimated errors, $\hat{\varepsilon}_i^2$, in place of the unknown $Var(\varepsilon_i)$. Hence, the formula for robust standard errors is given by:

$$\sqrt{\sum_{i=1}^{n}\left\{\left(\frac{(x_i - \bar{x})}{\sum(x_i - \bar{x})^2}\right)^2 \hat{\varepsilon}_i^2\right\}}$$

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Formulas for Calculating R^2

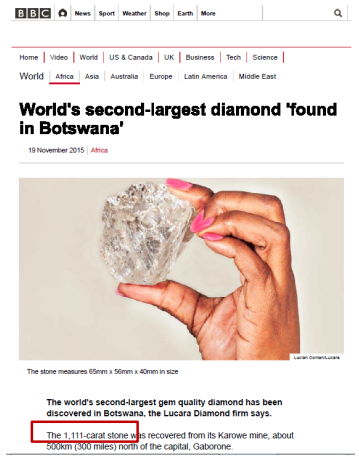
- R^2 Statistic Equations

In case you want to understand how R^2 and its components are calculated, here are the related equations:

$$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2,\ SSR = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2,\ SSE = \sum_{i=1}^{n} e_i^2,$$

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^{n} e_i^2,}{\sum_{i=1}^{n}(y_i - \bar{y})^2,}$$

World's Second-Largest Diamond Found



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World's second-largest diamond 'found in Botswana'

19 November 2015 | Africa

The stone measures 65mm x 56mm x 40mm in size

The world's second-largest gem quality diamond has been discovered in Botswana, the Lucara Diamond firm says.

The 1,111-carat stone was recovered from its Karowe mine, about 500km (300 miles) north of the capital, Gaborone.

Back